A7.3 Locked accesses

AXI4 does not support locked transactions. However, an AXI3 implementation must support locked transactions.

Note

AXI4 removes support for locked transactions because:

- the majority of components do not require locked transactions
- the implementation of locked transactions has a significant effect on:
  - the complexity of the interconnect
  - the ability to make QoS guarantees.

In this specification, **AxLOCK** indicates **ARLOCK** or **AWLOCK**.

When a master uses the **AxLOCK** signals for a transaction to show that it is a locked transaction then the interconnect must ensure that only that master can access the targeted slave region, until an unlocked transaction from the same master completes. An arbiter within the interconnect must enforce this restriction.

Before a master starts a locked sequence of either read or write transactions it must ensure that it has no other transactions waiting to complete.

Any transaction with **AxLOCK** indicating a locked transaction forces the interconnect to lock the following transaction. Therefore, a locked sequence must always complete with a final transaction that does not have **AxLOCK** indicating a locked transaction. This final transaction is included in the locked sequence and effectively removes the lock.

When completing a locked sequence, before issuing the final unlocking transaction, a master must ensure that all previous locked transactions are complete. It must then ensure that the final unlocking transaction has completed before it starts any further transactions.

The master must ensure that all transactions in a locked sequence have the same **AxID** value.

Note

Locked accesses require the interconnect to prevent any other transactions occurring while the locked sequence is in progress, and can therefore have an impact on the interconnect performance. This specification recommends that locked accesses are only used to support legacy devices.

This specification recommends the following restrictions, but they are not mandatory:

- keep any locked transaction sequence within a single 4KB address region
- limit any locked transaction sequence to two transactions.

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